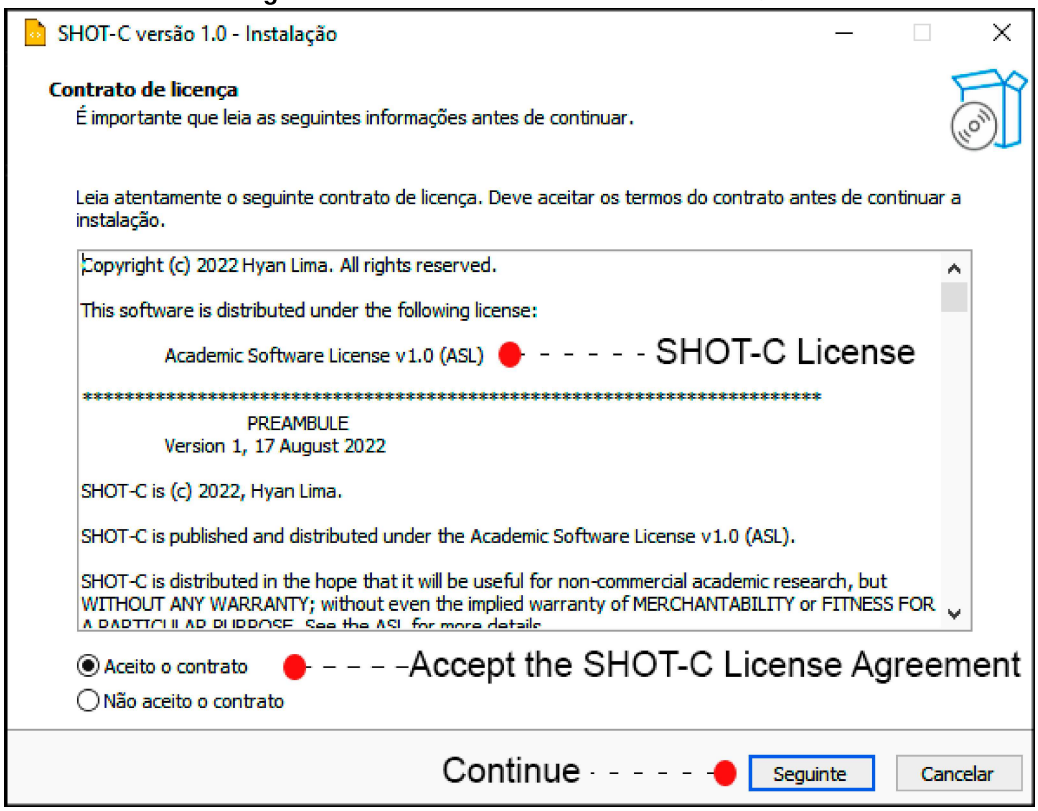
APPENDIX A – SHOT-C Software User Manual

A.1 Installing the SHOT-C Software Using the Installation Wizard

This manual has as reference the professional Windows 10 operating system version 21H2. After downloading SHOT-C from the Internet or building the executable in any way it suits, install the software using the executable file “SHOT-C Installer.exe”. Double-clicking this executable file brings up the user interface as shown in Figure 61. Throughout this manual, a signaling based on red circular markers was adopted, with a quota line and a larger-scale text, to draw attention to window elements, as well as to guide actions that need to be performed.

Figure 61 – SHOT-C Installation Window 01/05

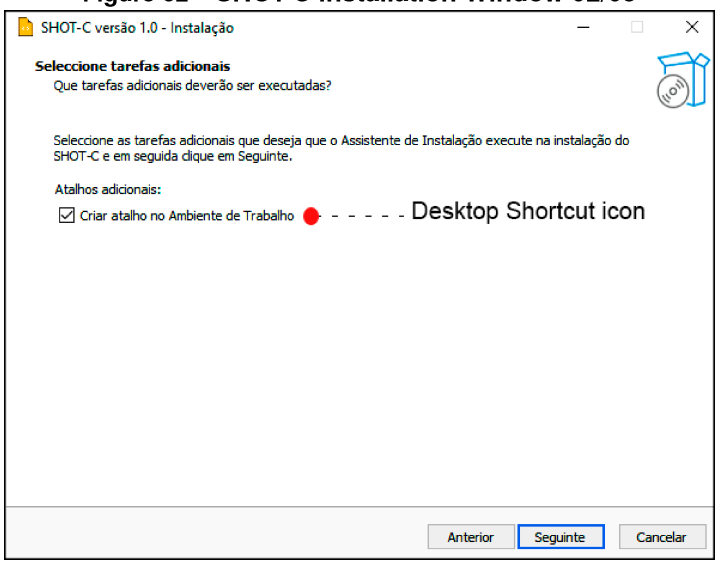


SOURCE: by the author (2022)

Click “I accept the agreement” and proceed to the next SHOT-C installation window by clicking the “Next” button. If a Windows 10 warning appears stating “Publisher: Unknown” arises by double clicking on the executable, ignore it and click “Yes” to allow the installation. In an unusual installation scenario, installation in admin mode may be required; to this end, on the executable and click “Execute as administrator”. If the installer still cannot be used, it is possible to create one using an installer creator. It is even possible to generate a SHOT-C executable via the Code::Blocks IDE, thereby eliminating the need for an installer. With regard to the construction of the SHOT-C source code in “.c” with TECGRAF libraries in the Code::Blocks IDE, carefully read the XML code available on the SHOT-C server.

Figure 62 shows the second installation window, where it is recommended to select (check) the box labeled “Create a shortcut on the desktop”; in other words, an icon will be created on the operating system’s desktop (Windows 10 desktop).

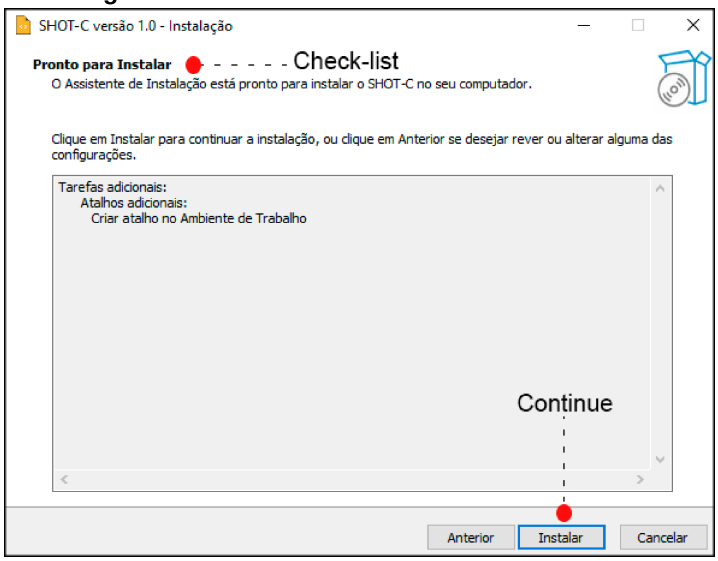
Figure 62 – SHOT-C Installation Window 02/05



SOURCE: by the author (2022)

Figure 63 shows the third installation window, which lists what will be installed on the operating system.

Figure 63 – SHOT-C Installation Window 03/05

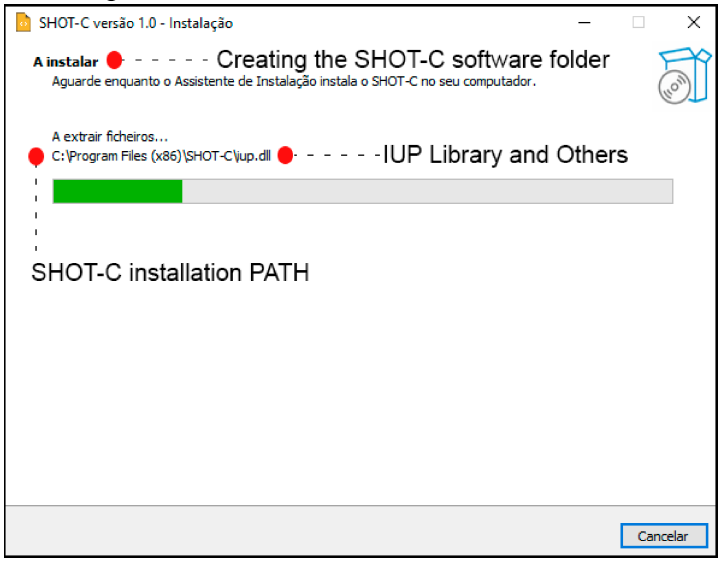


SOURCE: by the author (2022)

Figure 64 shows the fourth installation window, where the SHOT-C installation is processed, creating a folder through which the executable will have access to the TECGRAF li-

braries; in other words, the executable must be able to “see” the DLLs (.dll libraries), otherwise SHOT-C will not work.

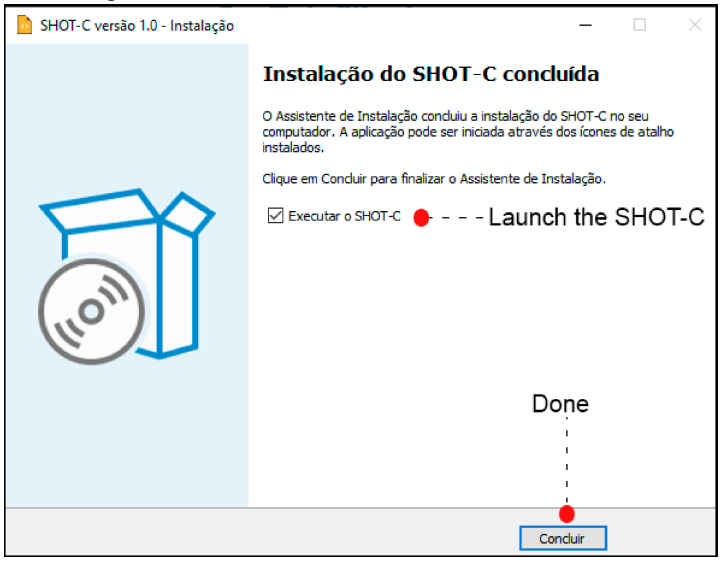
Figure 64 – SHOT-C Installation Window 04/05



SOURCE: by the author (2022)

Figure 65 shows the fifth, and last, installation window; after clicking “Finish”, SHOT-C starts up.

Figure 65 – SHOT-C Installation Window 05/05



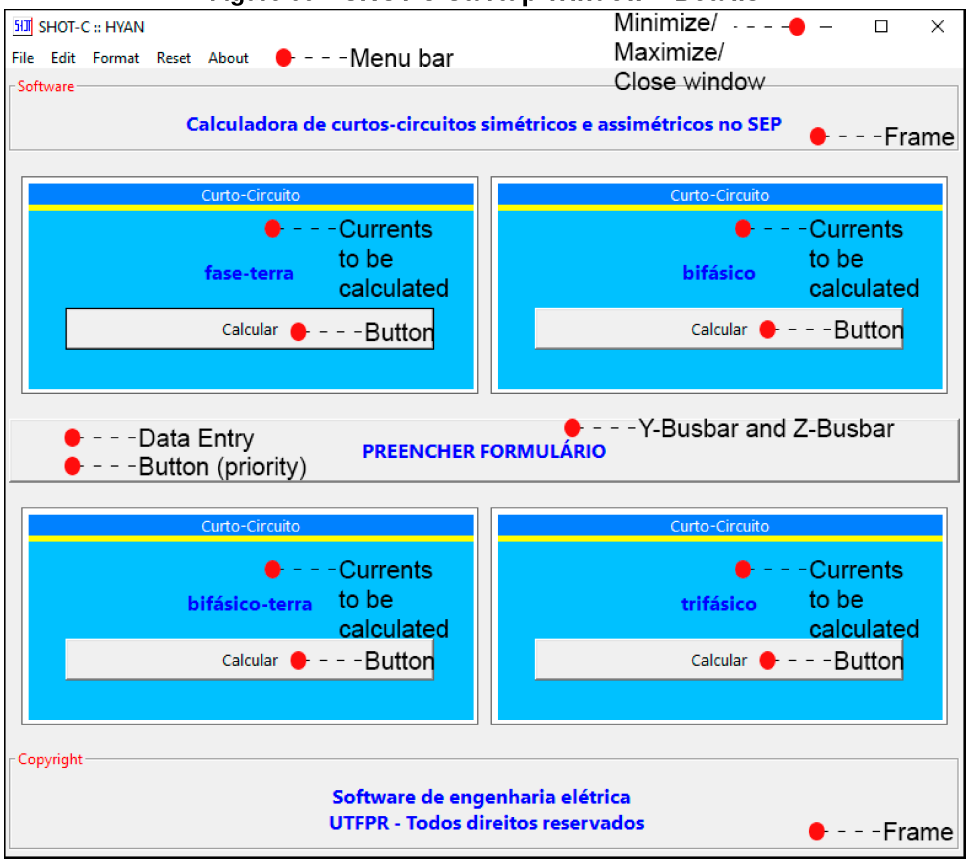
SOURCE: by the author (2022)

All five installation windows are automatically created by a computer program called “Inno Setup” (an installer creator), which was developed by Jordan Russell in 1997. This program is used to install other computer programs on the operating system; in other words, Inno Setup was used to install SHOT-C on the Windows 10 operating system.

A.2 Exploring the SHOT-C User Interface

Figure 66 shows SHOT-C's first graphical user interface, that is, the SHOT-C startup or SHOT-C's main window. In this figure, the window elements are highlighted with red circular markers, with a quota line and a larger text, to draw attention to the elements of the windows. It is important to emphasize that the User Form interface is a central element of SHOT-C, which is generated via the central button on the SHOT-C's main window; this button is the first one that must be clicked to enter data for the elements modeled for a short circuit.

Figure 66 – SHOT-C Startup Window – Details

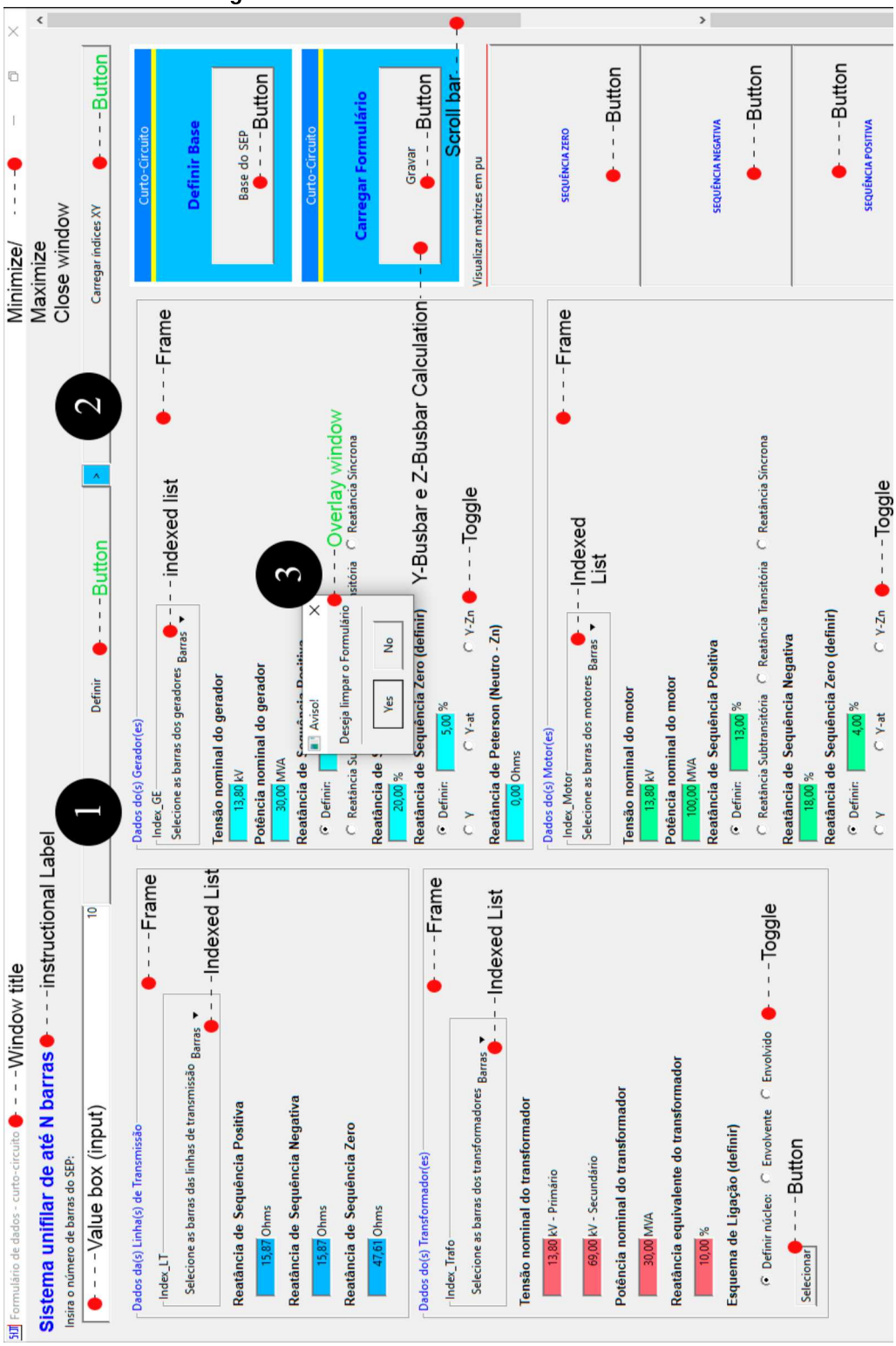


SOURCE: by the author (2022)

A Figura 67 mostra a segunda interface gráfica de usuário do SHOT-C, ou seja, o Formulário do SHOT-C. Nessa figura, além dos marcadores circulares vermelhos, também, está sendo sinalizado com três numeradores circulares, em preto e em grande escala, os três passos iniciais para uso do Formulário do SHOT-C.

Figure 67 shows the second graphical user interface of SHOT-C, namely, the SHOT-C's Form. In this figure, besides the red circular markers, three large black numbered circles highlight the three initial steps for using the SHOT-C Form.

Figure 67 – SHOT-C's Form Window – Details



SOURCE: by the author (2022)

The following is shown in Figure 68, this shows, once again, the second graphical user interface of SHOT-C, that is, the SHOT-C's Form, with step-by-step instructions on how to fill out the Form and calculate Y-Bus and Z-Bus. As mentioned earlier, the Form is used to input the specific values for elements modeled under short-circuit conditions. As stated previously,

the Form is where real values for the short-circuit modeled elements are to be entered; these elements include non-long transmission lines, synchronous generators, two-winding transformers, and synchronous motors. Loads with complex impedance may likewise be introduced as transmission lines.

The step-by-step instructions are indicated by colored arrows with larger text; just following the numbers in ascending order. Among these arrows, those with the same numbering indicate the possibility of choosing the EPS elements into which the real values are desired to be inserted. Even if, if the SHOT-C's Form fields are filled out, nothing will happen during the calculation process unless you click on the list index in the frame. It should be emphasized, therefore, that SHOT-C does not retrieve data from the form by clicking on "Load Form"; rather, it processes the real values stored in an indexed memory location; and these values, in turn, are retrieved from the Form by clicking on the index of the indexed list for the desired EPS element. The 7 steps that guide the use of the SHOT-C's Form are:

Step 1: After entering the number of busbars for the EPS (ring-type or radial), click the button next to the text box; that is, click on "Define" button adjacent in the text box.

Step 2: To trigger the indexing of the lists in all frames of the SHOT-C's Form, the "Load XY Indices" button must be clicked;

Step 3: SHOT-C asks whether the SHOT-C's Form should be cleared; the previous values are for example purposes only and can be reused, with the exception of the toggles;

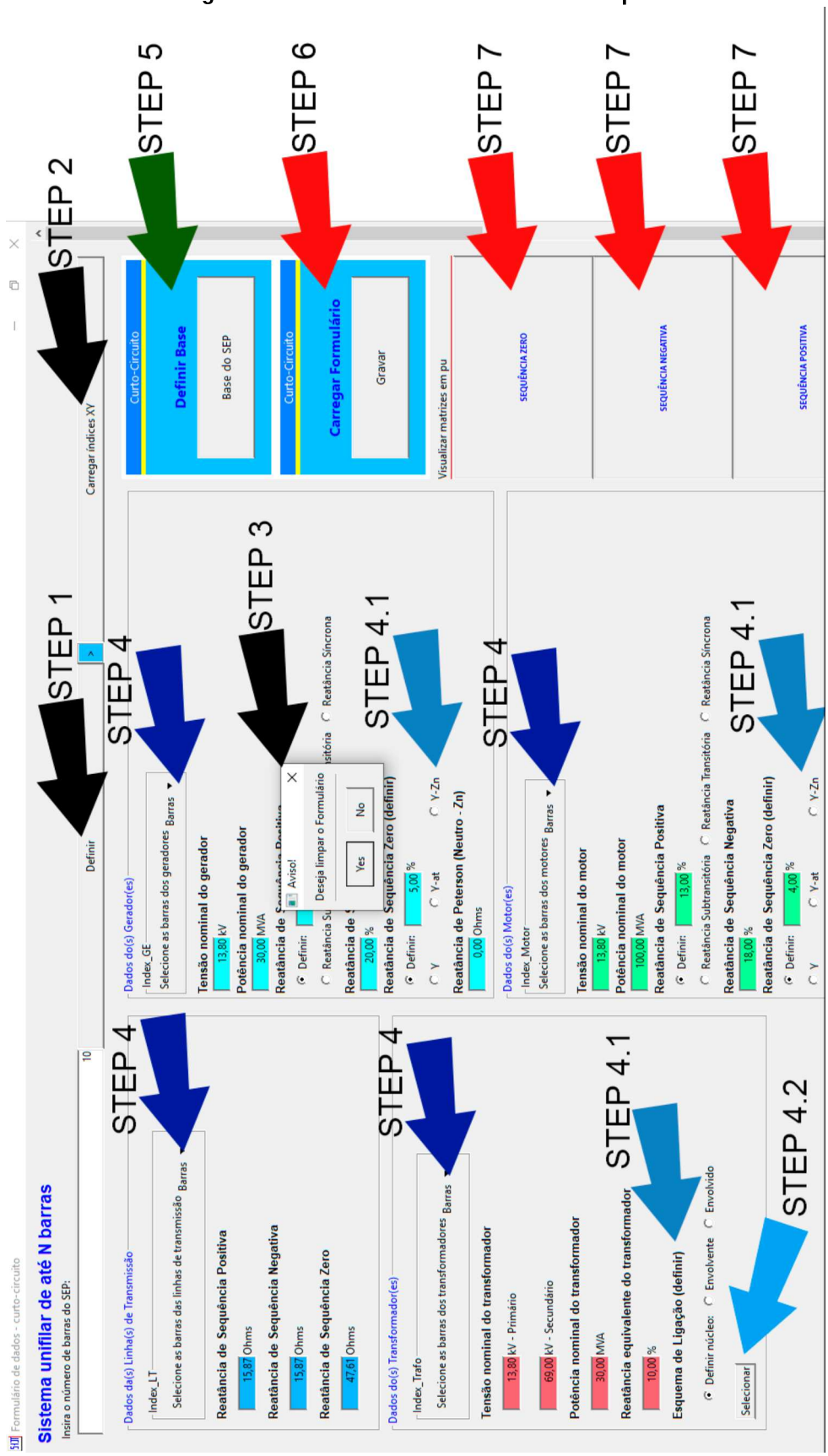
Step 4: When you click the down arrow, an indexed list appears. After entering the actual values for the EPS element, click the index corresponding to the busbar (X:X) to which the element is connected, or the index between busbars (X:Y or Y: X) of the element between busbars; all other indices not clicked will be processed by SHOT-C as nonexistent EPS elements;

Step 5: the "EPS Base" button is used to define a busbar on the side where the Base voltage and Base apparent power are located; any values may be entered;

Step 6: The "Load Form" button is used to process the calculation of all Bases, the conversion to pu for the EPS elements, and the calculation of Y-Bus and Z-Bus; ignore the error message for sequences that have no values entered;

Step 7: when clicking on the buttons "Zero Sequence", "Negative Sequence" and "Positive Sequence', the spreadsheets of Zpu, Ypu and Zth_pu are displayed, respectively, the spreadsheets of the pu matrices: of impedances of the elements of EPS modeled for short circuit, of admittances (Y-Bus) and Thévenin (Z-Bus).

Figure 68 – SHOT-C's Form Window – 7 steps



SOURCE: by the author (2022)

A.3 Practical Examples - 7-Step Guide

To clarify how to use SHOT-C, following are given some practical examples; although these examples have different characteristics, it is always possible to follow the 7 steps:

Example EX1: To insert a transmission line between busbars 7 and 8 of a 9-bus EPS ring-type, where only positive and negative sequences are calculated, the following steps are taken:

Step 1-EX1: Fill out the value box with number 9 and click on the button “Define”;

Step 2-EX1: Click the “Load XY Indices” button;

Step 3-EX1: SHOT-C asks whether the SHOT-C’s Form should be cleared; the previous values are for example purposes only and can be reused, with the exception of the toggles;

Step 4-EX1: First, enter the complex impedances without the imaginary part j and without the real part; commas are allowed (but decimal points are not allowed – otherwise, it must be rebuilt and compiled from source code); in this case, the impedances are the positive- and negative-sequence reactances (leave the zero-sequence reactance field blank or ignore it – since no index will be loaded). When clicking the downward-pointing arrow in the frame index_TL, an indexed list containing 72 elements X:Y or Y:X (are 9x9 = 81 indexes minus 9 X:X indexations of the main diagonal) is displayed. Thus, moving the scroll bar from the indexed list of the LT to element 7:8 is done by clicking this index. This automatically locks the 7:8 and 8:7 elements for the insertion of transformer or TL reactances ;

Step 5-EX1: It is recommended to insert at least two EPS elements (for N = 2 busbars), including the possibility of splitting the TL reactance into two equal parts (or any other ratio), adjusting the numbering of the single-line diagram under study. Thus, after filling out in all EPS elements, click the “Define Base” button and enter the Base’s voltage, Base’s apparent power, and the busbar on the Base’s side;

Step 6-EX1: If the entry was made correctly (careful attention is required, including filling out a check-list with a mechanical pencil on the diagram being studied), click the “Load Form” button. Several messages will appear regarding the Base’s voltage of the other busbars, along with progress messages; at the end, a completion message should appear, click “OK” on all messages. Ignore the message stating that the zero-sequence matrix cannot be inverted because the determinant is zero;

Step 7-EX1: After the completion warning, SHOT-C will have loaded a memory containing Y-Bus and Z-Bus; thus, it is possible to ignore this Step 7 and go directly to the initial window for calculating short-circuit currents, if desired. It is recommended, however, to click the

"Negative Sequence" and "Positive Sequence" buttons, corresponding to the sequences for which the calculation was intended; to verify $Zpu-$, $Ypu-$, and Zth_pu- for the negative sequence, and $Zpu+$, $Ypu+$, and Zth_pu+ for the positive sequence.

Example EX2: to insert two transmission lines, one between busbars 7:8 and another between busbars 1:2, in a radial-type EPS with 8 busbars, where only the positive sequence is intended to be calculated, the following steps are performed – The SHOT-C does not yet have a mechanism to recreate smaller index lists after inserting a larger index one, so it is necessary to restart SHOT-C by closing it and then double-clicking its executable icon on the desktop again – to obtain a properly indexed list:

Step 1-EX2: Fill out the value box with number 8 and click on the button "Define";

Step 2-EX2: Click the "Load XY Indices" button;

Step 3-EX2: SHOT-C asks whether the SHOT-C's Form should be cleared; the previous values are for example purposes only and can be reused, with the exception of the toggles;

Step 4-EX2: When clicking the downward-pointing arrow in the frame index_TL, an indexed list containing 56 X:Y or Y:X elements is displayed (there are 8x8 = 64 index entries minus 8 X:X entries on the main diagonal). Thus, by moving the scroll bar of the TL index list to the 1:2 element and then clicking on this index after entering the positive-sequence reactance value. Once again, by moving the scroll bar of the TL index list to the 7:8 element, and then clicking on this index after entering the positive-sequence reactance value. Consequently, elements 1:2, 2:1, 7:8, and 8:7 are automatically locked for the entry of transformer or TLs reactances;

Step 5-EX2: after filling in all elements of the EPS – it is not permitted to force open-circuit impedances; that is, if there are 8 defined busbars, there must be a single-line diagram with at least one combination connecting busbar 1 to busbar 8. Then, click the "Define Base" button and fill out the Base's voltage, Base's apparent power, and the Base's busbar on the Base's side; Several messages will appear regarding the Base's voltage of the other busbars, along with progress messages; at the end, a completion message should appear, click "OK" on all messages. Ignore the message stating that the zero-sequence matrix cannot be inverted because the determinant is zero;

Step 6-EX2: If the entry was made correctly (careful attention is required, including filling out a checklist with a mechanical pencil on the diagram being studied), click the "Load Form" button. Several messages will appear regarding the Base's voltage of the other busbars, along with progress messages; at the end, a completion message should appear, click "OK" on all messages. Ignore the warning about the impossibility of inversion of the negative and zero sequence matrices due to the determinant being null;

Step 7-EX2: After the completion warning, SHOT-C will have loaded a memory containing Y-Bus and Z-Bus; thus, it is possible to ignore this Step 7 and go directly to the initial window for calculating short-circuit currents, if desired. It is recommended, however, to click the "Positive Sequence" button, corresponding to the sequence for which the calculation was intended; to verify $Zpu+$, $Ypu+$, and Zth_pu+ for the positive sequence.

Example EX3: For a 4-bus radial-type EPS, insert: a shell-core transformer and star connection grounded at both the primary and secondary sides, between busbars 1 and 2; two transmission lines, one between busbars 2 and 3 and another between busbars 2 and 4; and finally, a synchronous generator at 1-bus and a synchronous motor at 3-bus. Where the positive, negative, and zero sequences are intended to be calculated, follow these steps – SHOT-C does not yet have a mechanism to recreate smaller lists after inserting a larger one, so you must restart SHOT-C by closing it and then double-clicking its icon on the desktop again:

Step 1-EX3: Fill out the value box with number 4 and click on the button "Define";

Step 2-EX3: Click the "Load XY Indices" button;

Step 3-EX3: SHOT-C asks whether the SHOT-C's Form should be cleared; the previous values are for example purposes only and can be reused, with the exception of the toggles;

Step 4-EX3: When clicking the downward-pointing arrow in the frame index_TL, an indexed list containing 12 X:Y or Y:X elements is displayed (there aare 4x4 = 16 index entries minus 4 X:X entries on the main diagonal). Thus, by moving the scroll bar of the TL index list to the 2:3 element and then clicking on this index after entering the reactance values. Once again, by moving the scroll bar of the LT index list to the 2:4 element, and then clicking on this index after entering the sequence reactance values. Consequently, elements 2:3, 3:2, 2:4, and 4:2 are automatically locked for the entry of transformer or TLs reactances; The next EPS element that can be entered (in no particular order) is the transformer; scrolling the Form's scroll bar highlights the frame for index_Trafo. Clicking the down arrow in that frame opens another indexed list containing 12 elements; proceed click on the 1:2 entry after entering the reactance values, nominal voltages and nominal power, besides selecting the type of core and the type of connection (as shown in the substeps 4.1 and 4.2 of the Figure 68 for transformers). Consequently, elements 1:2 and 2:1 are automatically locked to prevent the entry of reactances for another transformer or a transmission line. Finally, in the region of the Form intended for the synchronous motor (frame), click the down arrow in the frame labeled index_Motor, and thus an indexed list is displayed containing 4 X:X elements (4 X:X indexations of the main diagonal); locate the 3:3 index and click on it. Similarly, in the region of the Form intended for the synchronous generator (frame), click the down arrow next to the frame index_GE, which displays an indexed list containing 4 X:X entries (4 X:X indices of the main diagonal); locate the 1:1 index and click on it.

Synchronous machines shall have their indexes, in the order of the 3:3 and 1:1; these indices will be locked to prevent the addition of any other synchronous machine; these indices should only be selected after entering the reactance values, nominal voltages and nominal powers, besides selecting the type of connection (as shown in the substeps 4.1 of Figure 68 for synchronous machines);

Step 5-EX3: After filling in all the elements of the EPS, click the “Define Base” button and enter the Base’s voltage, Base’s power, and the busbar on the Base’s side;

Step 6-EX3: If the data entry was made correctly (care is required, and it is recommended to use a pencil to mark a checklist on the diagram being studied), click the “Load Form” button; several messages regarding the Base’s voltage of the other busbars and progress messages will appear; at the end, a completion message should appear, then click “OK” on all messages;

Step 7-EX3: After the completion warning, SHOT-C will have loaded a memory containing Y-Bus and Z-Bus; thus, it is possible to ignore this Step 7 and go directly to the initial window for calculating short-circuit currents, if desired. It is recommended, however, to click the “Zero Sequence”, “Negative Sequence”, and “Positive Sequence” buttons, corresponding to the sequence for which the calculation was intended; to verify $Zpu0$, $Ypu0$, and Zth_pu0 for zero sequence; $Zpu-$, $Ypu-$, and Zth_pu- for negative sequence; and $Zpu+$, $Ypu+$, and Zth_pu+ for positive sequence.

These three examples alone provide a general overview of SHOT-C’s usability, with two addenda:

- SHOT-C’s graphical interfaces were built to be centralized windows; that is, since by minimizing the initial window and then maximizing it again, while multiple SHOT-C windows are open, then all windows will overlap after maximization, requiring dragging the windows with the mouse to see which are open;
- It is not necessary to reopen the transformer connection window for each new transformer, since, when selecting the connection type, the number of that connection is temporarily stored until the desired index is clicked; thus, when that index is clicked, the connection number is loaded.

THE END.